

Unit 15

HALOGEN COMPOUNDS

33 items · 70 marks

Syllabus sections in this unit

15.1 Halogenoalkanes

Unit 15: Halogen compounds

1. M/J 26/P21/Q5(e)

Context

- 5 Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

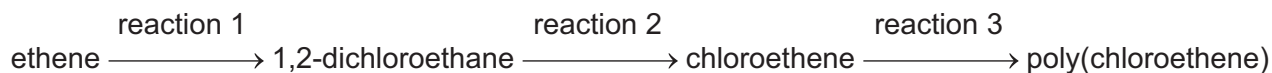


Figure 5.1

- (e) When 1,2-dichloroethane is heated in reaction 2, thermal decomposition occurs. The reaction is endothermic.

State the type of reaction that occurs.

[1]

2. MAR 26/P22/Q3(a)

Context

- 3 Secondary halogenoalkanes are important building blocks in organic chemistry. 2-chlorobutane is an example of a secondary halogenoalkane.

2-chlorobutane

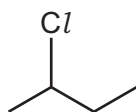


Figure 3.1

- (a) Explain why 2-chlorobutane is classified as a secondary halogenoalkane.

[1]

3. MAR 26/P22/Q3(d)

Context

3 Secondary halogenoalkanes are important building blocks in organic chemistry.

2-chlorobutane is an example of a secondary halogenoalkane.

2-chlorobutane

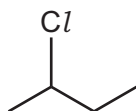


Figure 3.1

(d) The rates of hydrolysis of three secondary halogenoalkanes, **B**, **C** and **D**, are compared.

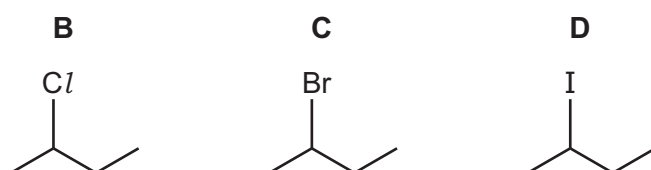


Figure 3.3

Each halogenoalkane is dissolved in ethanol and reacted with $\text{AgNO}_3(\text{aq})$.

Identify the order of the rates of hydrolysis of compounds **B**, **C** and **D**. Explain your answer.

fastest > > slowest

explanation

[2]

4. MAR 26/P22/Q4(c)

Context

4 Cyanoethanoic acid, $\text{CH}_2(\text{CN})\text{COOH}$, is a weak organic acid.

(c) Figure 4.3 shows a synthesis of $\text{CH}_2(\text{CN})\text{COOH}$.

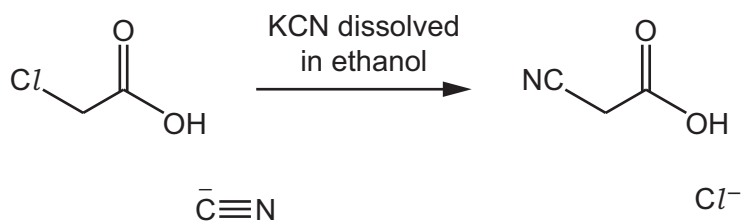


Figure 4.3

(i) The reaction mechanism of the synthesis in Figure 4.3 has a single step.

Complete Figure 4.3 to show this mechanism. Include lone pairs, charges, dipoles and curly arrows as appropriate.

[2]

(ii) Identify the nucleophile in Figure 4.3.

[1]

(iii) Identify the mechanism of the reaction in Figure 4.3.

[1]

5. O/N 25/P22/Q4(b)

Context

4 Fig. 4.1 shows how propane, C_3H_8 , can be converted to propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$.

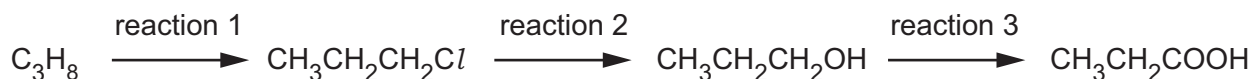


Fig. 4.1

(b) State the reagent and solvent required for reaction 2.

[1]

6. O/N 25/P24/Q3(b)(ii)-(iii)

Context

- 3 Propan-2-ol, $(\text{CH}_3)_2\text{CHOH}$, is sometimes added to fuel to help it burn.

Fig. 3.1 shows some reactions of propan-2-ol.

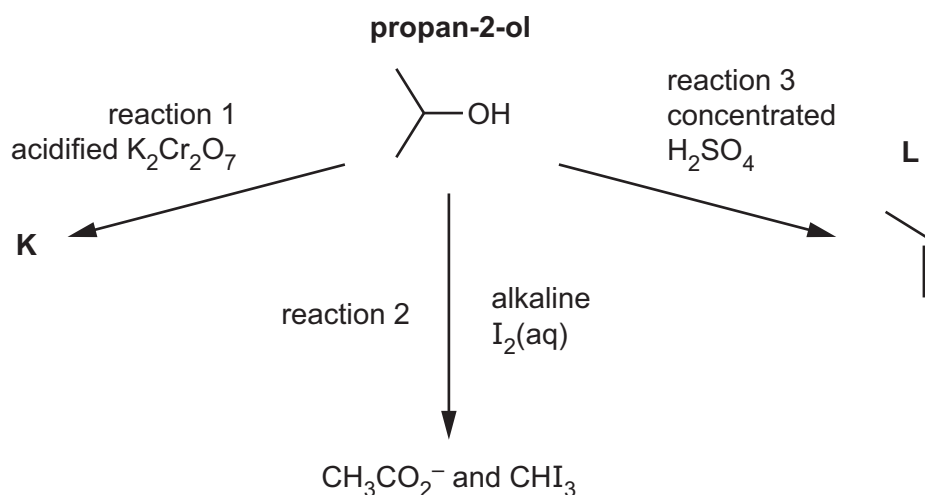


Fig. 3.1

Context: part (b) (introduction)

- (b) Propan-2-ol reacts with sodium to produce $(\text{CH}_3)_2\text{CH-O}^-$ anions.

These anions react with 2-bromopropane to form compound **N**, as shown in Fig. 3.3.

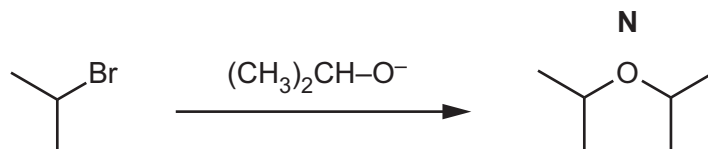


Fig. 3.3

- (ii) The reaction of 2-bromopropane with $(\text{CH}_3)_2\text{CH-O}^-$ anions follows an $\text{S}_{\text{N}}1$ mechanism.

Complete Fig. 3.4 to show this mechanism. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 3.4

- (iii) Suggest how the rate of the $\text{S}_{\text{N}}1$ reaction would change, if at all, if 2-chloropropane were used instead of 2-bromopropane.

Explain your answer.

[3]

[2]

7. O/N 25/P24/Q4(b)(i)

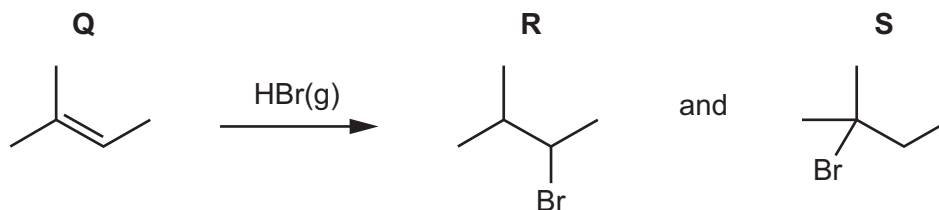
Context

4 Compounds **P** and **Q** are structural isomers.

**Fig. 4.1**

Context: part (a)(v)

(v) **Q** reacts with HBr(g) to produce two structural isomers, **R** and **S**, as shown in Fig. 4.2.

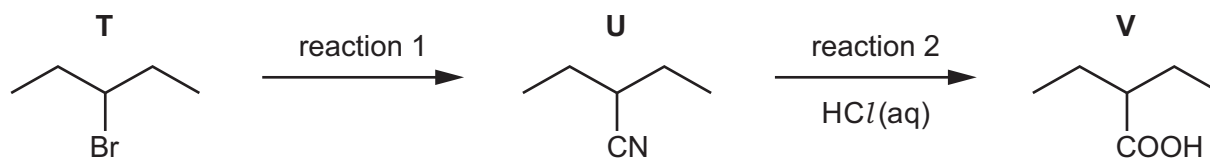
**Fig. 4.2**

State and explain why isomer **S** is the major product of the reaction.

[2]

Context: part (b) (introduction)

(b) Fig. 4.3 shows a synthesis starting from **T**, a different isomer of **R** and **S**.

**Fig. 4.3**

(i) Identify the reagent and conditions for reaction 1.

[1]

8. M/J 25/P22/Q4(b)(ii)

Context: part (b) (introduction)

(b) Propan-1-ol is converted to compound **T** in a three-step synthesis.

In step 1, SOCl_2 is added to propan-1-ol to produce compound **E**.

In step 2, **E** reacts with a suitable reagent to produce butanenitrile.

In step 3, butanenitrile is heated with NaOH(aq) .

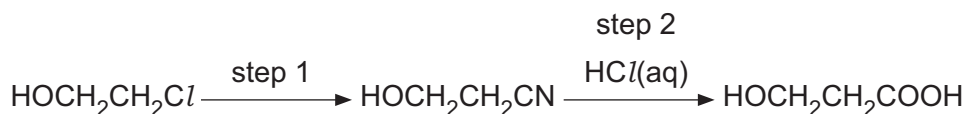
(ii) Identify the reagent and conditions required in step 2.

[1]

9. M/J 25/P23/Q6(c)(i)

Context: part (c) (introduction)

- (c) $\text{HOCH}_2\text{CH}_2\text{Cl}$ reacts in a two-step synthesis to produce $\text{HOCH}_2\text{CH}_2\text{COOH}$, as shown in Fig. 6.2.

**Fig. 6.2**

- (i) State the reagent and conditions for step 1.

[1]

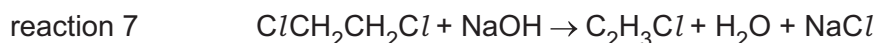
10. MAR 25/P22/Q1(c)(ii)

Context

- 1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

Context: part (c) (introduction)

- (c) 1,2-dichloroethane, $\text{ClCH}_2\text{CH}_2\text{Cl}$, reacts with NaOH to produce an unsaturated compound, $\text{C}_2\text{H}_3\text{Cl}$, as shown in reaction 7.



- (ii) State the conditions for reaction 7.

[1]

11. O/N 24/P21/Q2(c)(iv)

Context

- 2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (c) (introduction)

- (c) CHCl_3 is another halogenoalkane. CHCl_3 forms when propanone reacts with NaClO .

NaClO is made from chlorine in a disproportionation reaction.

- (iv) Aqueous AgNO_3 dissolved in ethanol reacts with an aqueous solution of CHCl_3 .

State what is observed in this reaction. Explain your answer.

[2]

12. O/N 24/P23/Q2(c)(iv)

Context

- 2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (c) (introduction)

- (c) CHCl_3 is another halogenoalkane. CHCl_3 forms when propanone reacts with NaClO .

NaClO is made from chlorine in a disproportionation reaction.

- (iv) Aqueous AgNO_3 dissolved in ethanol reacts with an aqueous solution of CHCl_3 .

State what is observed in this reaction. Explain your answer.

[2]

13. M/J 24/P22/Q4(b)

Context

4 $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ exists as a pair of stereoisomers.

(b) A sample of $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ reacts with NaOH to make $\text{CH}_3(\text{CH}_2)_5\text{CH(OH)CH}_3$ in an $\text{S}_{\text{N}}1$ mechanism.

Complete Fig. 4.1 to show the mechanism for the reaction of $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ and NaOH .

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Fig. 4.1**

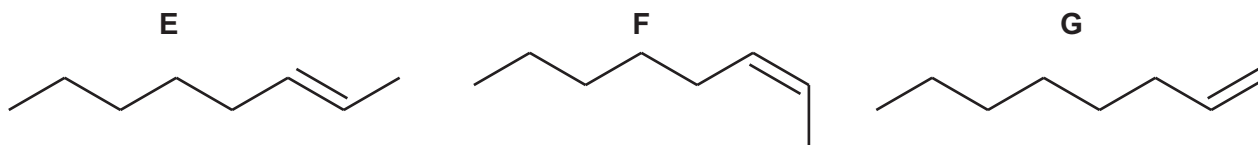
[3]

14. M/J 24/P22/Q4(d)

Context

4 $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ exists as a pair of stereoisomers.

(d) $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ is heated with **D** to produce three different molecules, **E**, **F** and **G**.

**Fig. 4.2**

(i) Name the type of reaction.

[1]

(ii) Identify **D** and the conditions used.

[1]

15. M/J 24/P23/Q4(c)

(c) Substitution reactions of NH_3 and OH^- with halogenoalkanes both involve a lone pair of electrons.

(i) Name the role of NH_3 and OH^- in these reactions.

[1]

(ii) Suggest which species, NH_3 or OH^- , is more reactive during these reactions. Explain your answer.

[1]

16. M/J 24/P23/Q4(d)

- (d) When 2-bromo-2-methylpropane reacts with OH^- , two mechanisms, $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$, both occur. The $\text{S}_{\text{N}}2$ mechanism has a slower rate.

Fig. 4.1 shows the reaction pathway diagram for the $\text{S}_{\text{N}}1$ mechanism.

Sketch a graph on Fig. 4.1 to show the reaction pathway for the $\text{S}_{\text{N}}2$ mechanism.

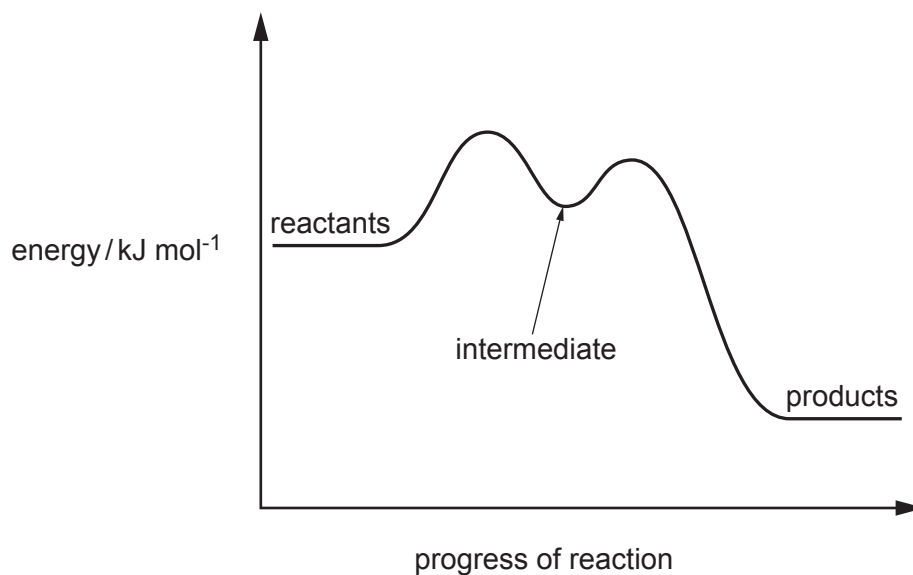
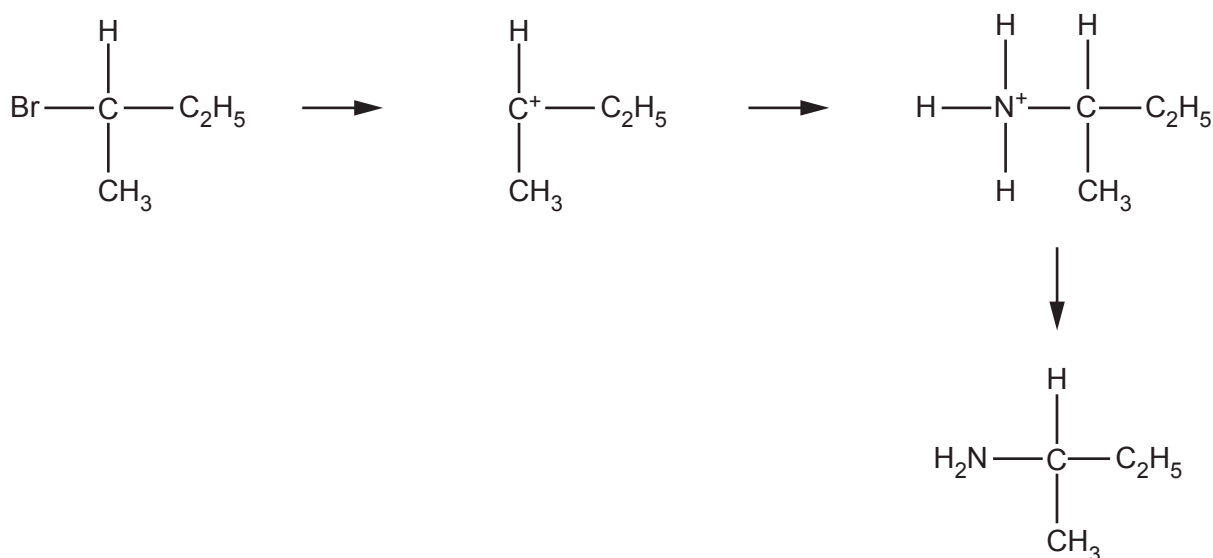


Fig. 4.1

[2]

17. M/J 24/P23/Q4(e)(i)-(ii)

- (e) (i) Complete Fig. 4.2 to show the mechanism for the S_N1 reaction that occurs when $\text{CH}_3\text{CHBrC}_2\text{H}_5$ reacts with NH_3 to produce $\text{CH}_3\text{CH}(\text{NH}_2)\text{C}_2\text{H}_5$. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Fig. 4.2**

[3]

- (ii) Identify the inorganic product that forms in the reaction in Fig. 4.2.

[1]

18. M/J 24/P23/Q4(f)

- (f) (i) Complete Table 4.1 by drawing the structural formula of the intermediate that is formed when 2-bromo-2-methylpropane reacts in an S_N1 reaction.

Table 4.1

	2-bromobutane	2-bromo-2-methylpropane
structural formula of intermediate in S_N1 reaction	$\begin{array}{c} \text{H} \\ \\ \text{C}^+ - \text{C}_2\text{H}_5 \\ \\ \text{CH}_3 \end{array}$	

[1]

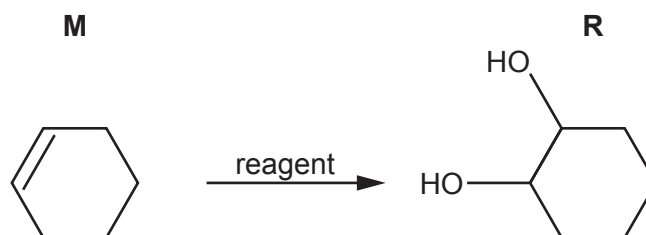
- (ii) Identify the halogenoalkane in Table 4.1 that has the greater tendency to react using the S_N1 mechanism. Explain your answer.

[2]

19. M/J 24/P23/Q5(b)(i)

Context: part (a)

- 5 (a)** **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

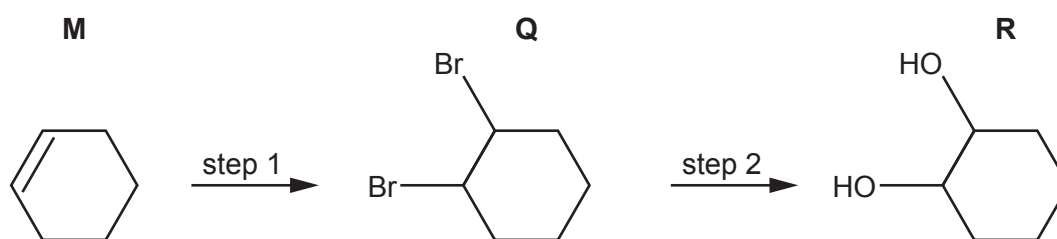
**Fig. 5.1**

Identify the reagent and conditions for this reaction.

[1]

Context: part (b) (introduction)

- (b)** **R** is also made from **M** by two steps, as shown in Fig. 5.2.

**Fig. 5.2**

- (i)** Identify the reagents and conditions for steps 1 and 2 in Fig. 5.2.

step 1

step 2

[2]

20. MAR 24/P22/Q2(d)(ii)

Context

2 Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

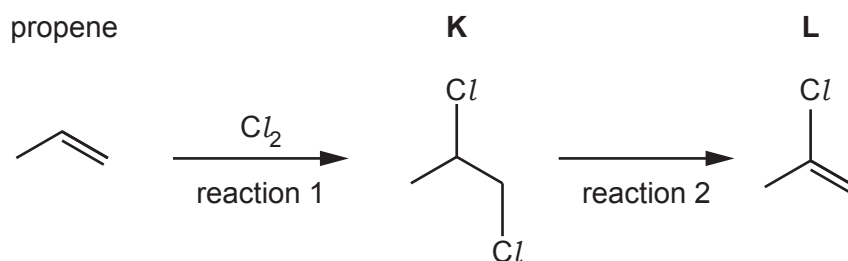
Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

Context: part (d) (introduction)

(d) Cl_2 reacts readily with propene to form **K**, 1,2-dichloropropane.

K can be used to form **L**.

**Fig. 2.1**

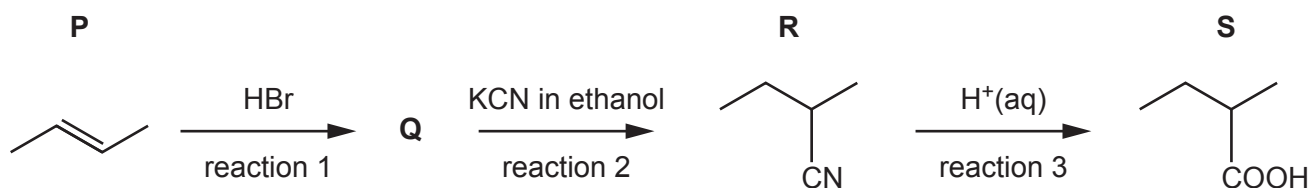
(ii) Identify the reagent and conditions for reaction 2.

[1]

21. MAR 24/P22/Q4(b)(ii)

Context

- 4** Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

**Fig. 4.1**

- (ii) Name the mechanism in reaction 2.

[1]

22. M/J 23/P21/Q3(e)

Context

- 3** Chlorine is a very reactive element.

- (e) CHCl_3 and HF are used to form CHClF_2 in a substitution reaction.

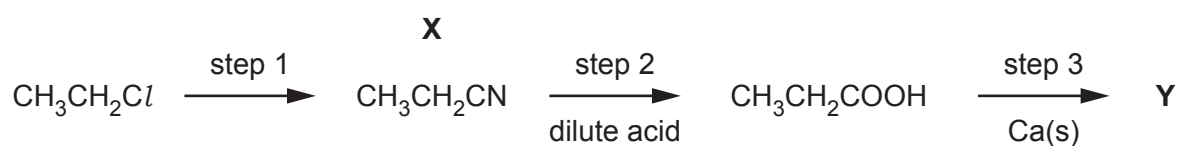
Construct an equation for this reaction.

[1]

23. M/J 23/P22/Q3(a)(i)

Context

- 3** Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from $\text{CH}_3\text{CH}_2\text{Cl}$.

**Fig. 3.1**

- (a) (i) State the reagent and conditions for step 1 in Fig. 3.1.

[1]

24. MAR 23/P22/Q4(a)(i)

Context

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

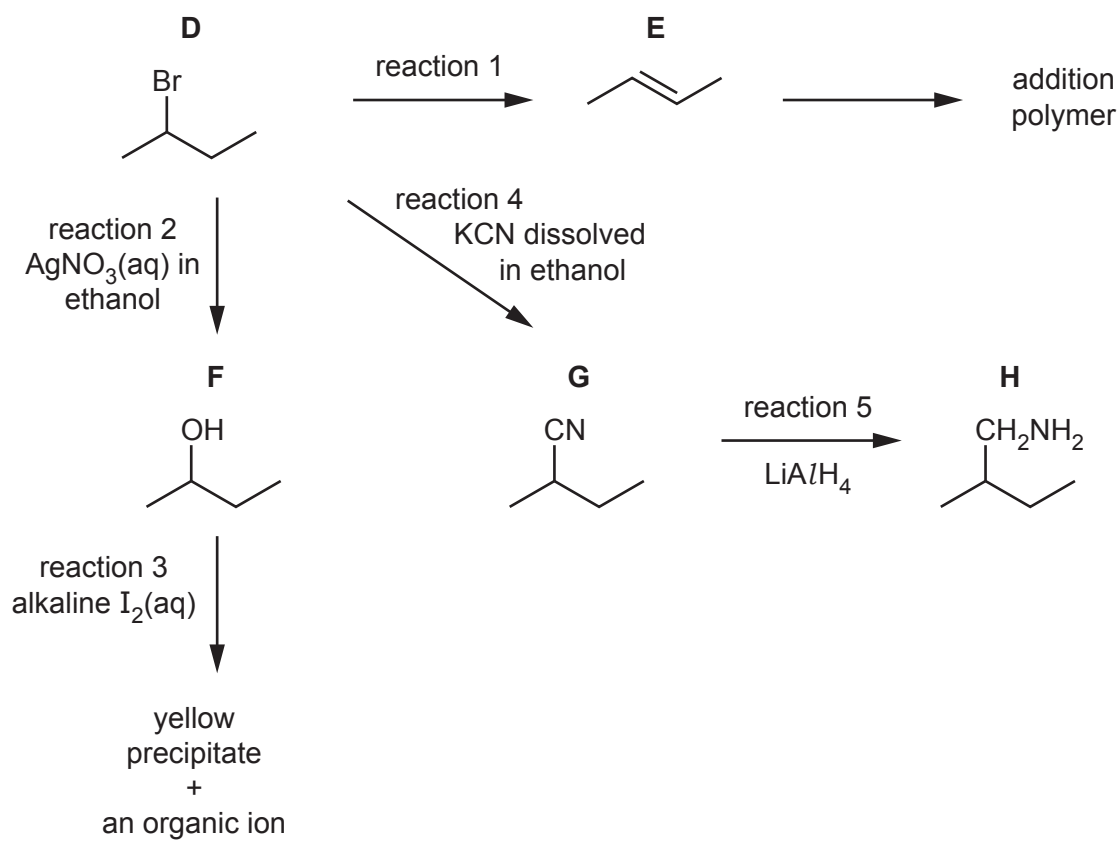


Fig. 4.1

(a) (i) State the reagent and conditions used to form **E** in reaction 1.

[1]

25. MAR 23/P22/Q4(b)(i)

Context

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

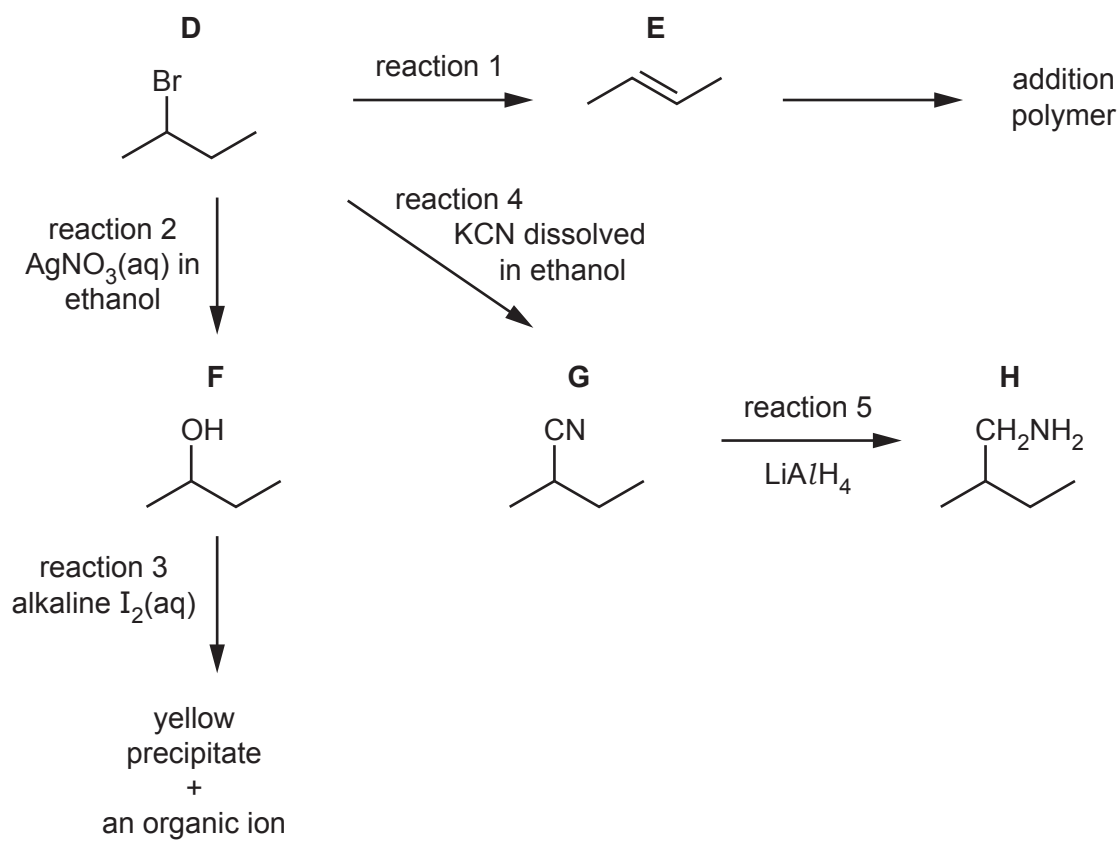


Fig. 4.1

(b) (i) Predict what is observed in reaction 2.

[1]

26. MAR 23/P22/Q4(c)(i)

Context

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

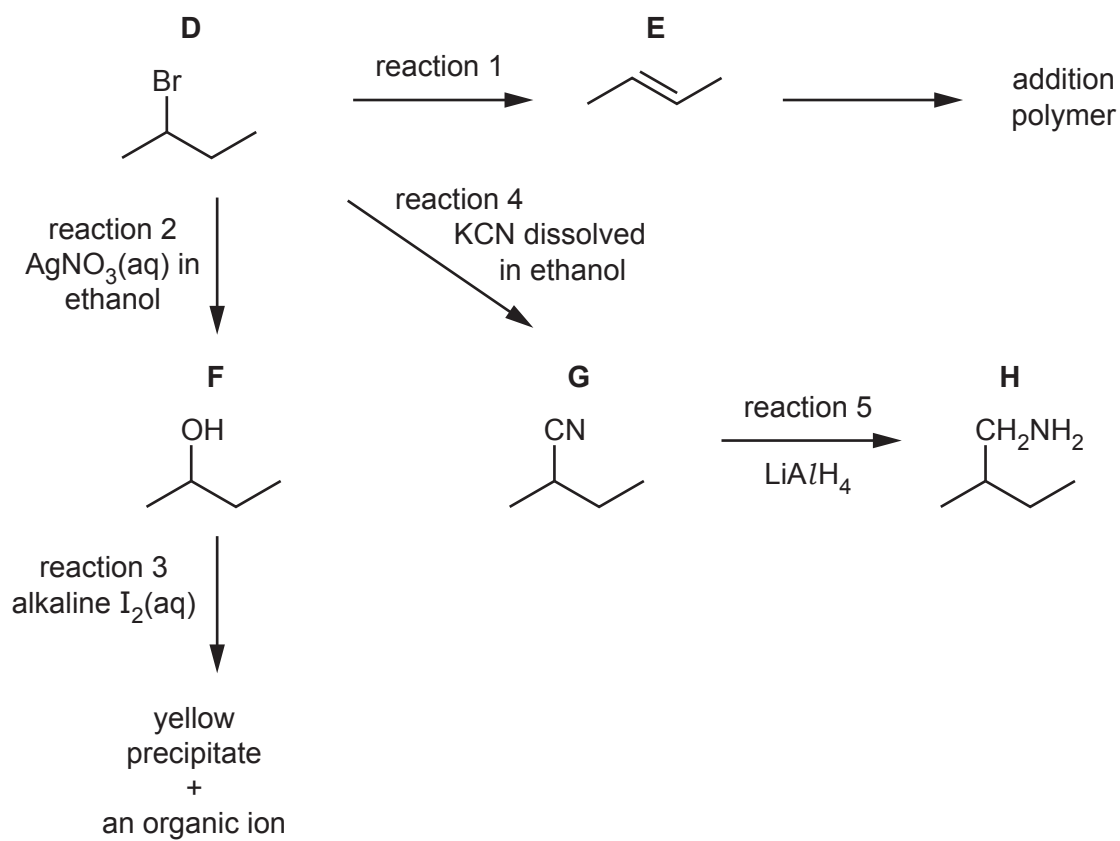


Fig. 4.1

(c) (i) State the type of reaction that occurs in reaction 4.

[1]

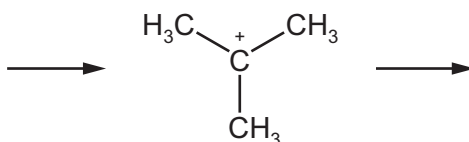
27. O/N 22/P22/Q1(e)

Context

- 1** Species such as NH_4^+ , CO_3^{2-} and PO_4^{3-} are examples of molecular ions.

- (e) $\text{OH}^-(\text{aq})$ reacts with 2-bromo-2-methylpropane in an $\text{S}_{\text{N}}1$ reaction. The molecular ion $(\text{CH}_3)_3\text{C}^+$ forms as the intermediate in this reaction.

- (i) Draw the mechanism for the S_N1 reaction of OH^- with 2-bromo-2-methylpropane. Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate. Draw the structures of the organic reactant and organic product.



[3]

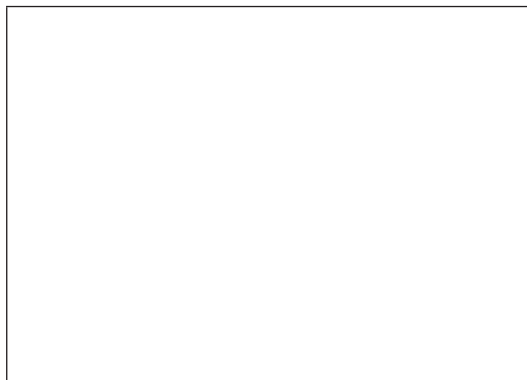
- (ii) 2-bromo-2-methylpropane is a tertiary bromoalkane.

Define tertiary bromoalkane.

[1]

- (iii) Organic compound **M** forms when 2-bromo-2-methylpropane is heated with **ethanolic** OH^- .

Draw the structure of **M**.

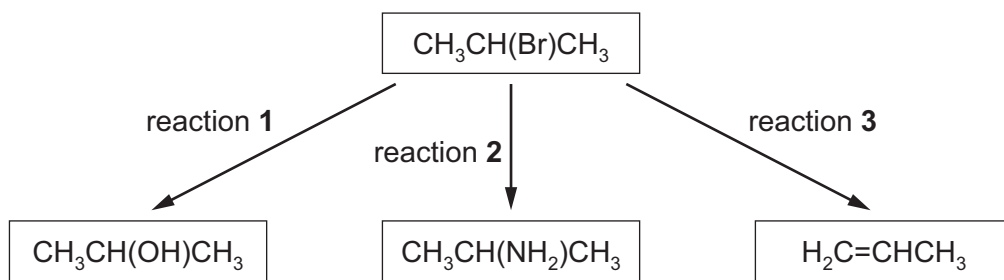


[1]

28. M/J 22/P22/Q5(a)

Context

5 Fig. 5.1 shows three reactions of 2-bromopropane, $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$.

**Fig. 5.1**

(a) Complete Table 5.1 for each reaction, by:

- stating the reagent and conditions used
- identifying the type of reaction that occurs.

Table 5.1

reaction	reagent and conditions	type of reaction
1		
2		
3		

[6]

29. M/J 22/P22/Q5(b)

Context

5 Fig. 5.1 shows three reactions of 2-bromopropane, $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$.

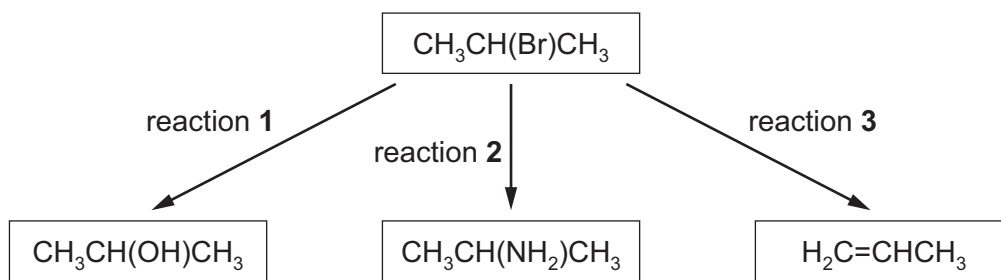


Fig. 5.1

(b) A sample of 2-iodopropane, $\text{CH}_3\text{CH}(\text{I})\text{CH}_3$, reacts under the same conditions as reaction 1 to produce $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$.

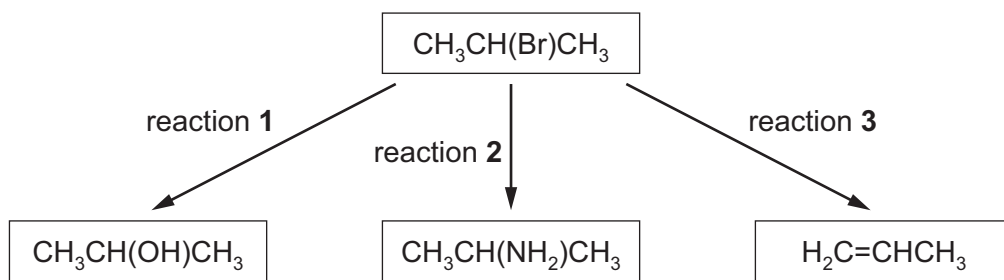
Explain why 2-iodopropane reacts at a faster rate than 2-bromopropane.

[2]

30. M/J 22/P22/Q5(c)(i)

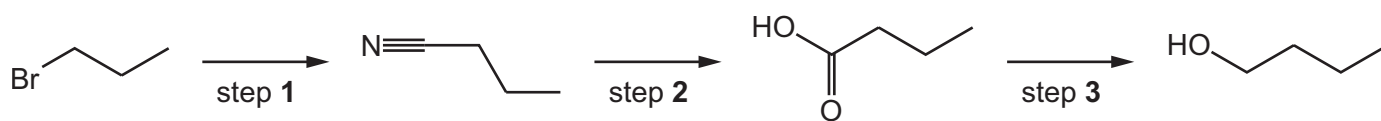
Context

5 Fig. 5.1 shows three reactions of 2-bromopropane, $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$.

**Fig. 5.1**

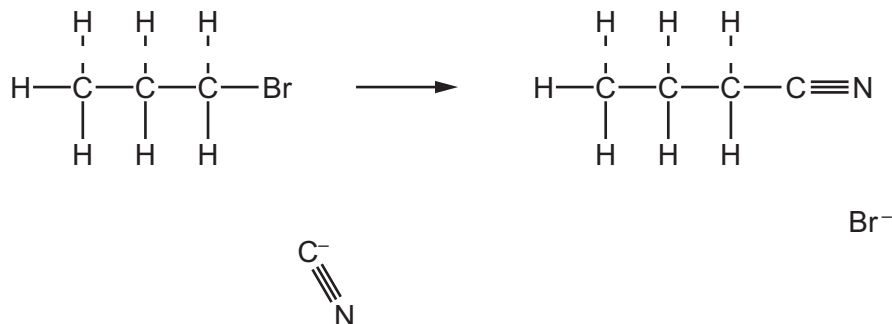
Context: part (c) (introduction)

(c) Fig. 5.2 shows how butan-1-ol can be made from 1-bromopropane in three steps.

**Fig. 5.2**

(i) In step 1, 1-bromopropane reacts with CN^- to form butanenitrile.

Complete Fig. 5.3 to show the mechanism for step 1. Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.

**Fig. 5.3**

[2]

31. M/J 22/P23/Q4(b)(i)-(iii)

Context: part (b) (introduction)

- (b) Two bottles labelled **Q** and **M** each contain a straight-chain halogenoalkane with molecular formula C_4H_9X , where **X** represents Cl, Br or I.
A sample from each bottle is added to separate samples of equal amounts of aqueous silver nitrate in ethanol. In each reaction, the same organic product, **T**, and a precipitate are made, as shown in Fig. 4.2.

**Fig. 4.2**

Table 4.1 describes the colour of each of the precipitates made.

Table 4.1

halogenoalkane added to $\text{AgNO}_3(\text{aq})$ in ethanol	colour of precipitate
Q	white
M	yellow

- (i) Identify the functional group present in **T** and name the type of reaction that occurs using the information in Fig. 4.2 and Table 4.1.

functional group in **T**

type of reaction

[2]

- (ii) Construct an ionic equation to describe the formation of the yellow precipitate produced when **M** reacts with $\text{AgNO}_3(\text{aq})$ in ethanol.

[1]

- (iii) Describe which reagent, **Q** or **M**, will produce a precipitate more quickly when each is added to $\text{AgNO}_3(\text{aq})$ in ethanol. Explain your answer.

reagent

[1]

32. M/J 22/P23/Q4(b)(v)

Context: part (b) (introduction)

- (b) Two bottles labelled **Q** and **M** each contain a straight-chain halogenoalkane with molecular formula C_4H_9X , where **X** represents Cl, Br or I.
A sample from each bottle is added to separate samples of equal amounts of aqueous silver nitrate in ethanol. In each reaction, the same organic product, **T**, and a precipitate are made, as shown in Fig. 4.2.

**Fig. 4.2**

Table 4.1 describes the colour of each of the precipitates made.

Table 4.1

halogenoalkane added to $\text{AgNO}_3(\text{aq})$ in ethanol	colour of precipitate
Q	white
M	yellow

- (v) Deduce the structure of the straight-chain halogenoalkane **M**.

[2]

33. MAR 22/P22/Q3(e)(iii)

Context

- 3** The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.

Context: part (e) (introduction)

- (e) HBr reacts with propene to form two bromoalkanes, $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ and $(\text{CH}_3)_2\text{CHBr}$.

- (iii) The reaction of $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ and NaOH is different depending on whether water or ethanol is used as a solvent.

Complete Table 3.2 to identify the organic and inorganic products of the reaction of $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ and NaOH in each solvent.

Table 3.2

solvent	organic product(s)	inorganic product(s)
water		
ethanol		

[2]

Answers Section

1. M/J 26/P21/Q5(e)

5(e)	elimination	1
------	-------------	---

2. MAR 26/P22/Q3(a)

3(a)	(organic molecule where) carbon (of C—Hal) is bonded to two other C atoms	1
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3. MAR 26/P22/Q3(d)

3(d)	M1 fastest D > C > B slowest M2 carbon—halogen gets weaker down the group (so breaks faster)	2
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4. MAR 26/P22/Q4(c)

4(c)(i)	M1 curly arrow from lone pair on C of CN ⁻ to ^{δ+} C of C—Cl M2 correct dipole of ^{δ+} C—Cl ^{δ-} AND curly arrow from bond to Cl	2
4(c)(ii)	CN ⁻ / cyanide	1
4(c)(iii)	nucleophilic substitution / S _N (2)	1

5. O/N 25/P22/Q4(b)

4(b)	sodium hydroxide AND water	1
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6. O/N 25/P24/Q3(b)(ii)-(iii)

3(b)(ii)	M1 correct dipole on ^{δ+} C—Br ^{δ-} AND curly arrow from C—Br bond to Br  M2 intermediate M3 curly arrow from lone pair of (CH ₃) ₂ CHO ⁻ to C ⁺	3
3(b)(iii)	M1 rate would decrease M2 C—Br is weaker (than C—Cl, so breaks more easily)	2

7. O/N 25/P24/Q4(b)(i)

4(b)(i)	KCN in ethanol (and heat)	1
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Answer for context part (a)(v)

4(a)(v)	M1 (reaction proceeds via a) more stable intermediate / carbocation M2 because of greater positive inductive effect of more alkyl groups	2
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8. M/J 25/P22/Q4(b)(ii)

4(b)(ii)	KCN / potassium cyanide in ethanol / C ₂ H ₅ OH AND heat	1
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9. M/J 25/P23/Q6(c)(i)

6(c)(i)	KCN/potassium cyanide in ethanol AND heat	1
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10. MAR 25/P22/Q1(c)(ii)

1(c)(ii)	ethanol (solvent) AND heat	1
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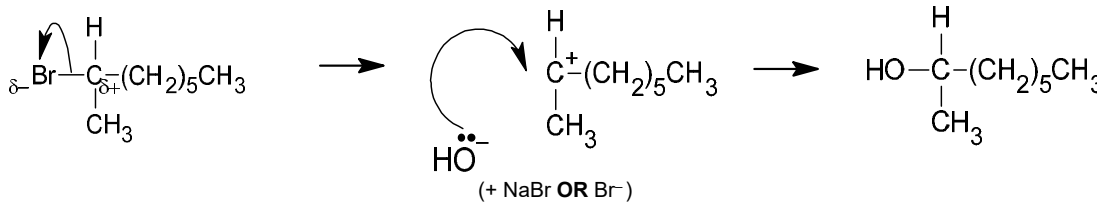
11. O/N 24/P21/Q2(c)(iv)

2(c)(iv)	M1 white precipitate/white ppt forms	1
	M2 CHCl_3 is hydrolysed / chloride ions are released OR Cl^- / chloride ions which are released react with Ag^+ OR silver chloride / AgCl is formed	1

12. O/N 24/P23/Q2(c)(iv)

2(c)(iv)	M1 white precipitate/white ppt forms	1
	M2 CHCl_3 is hydrolysed / chloride ions are released OR Cl^- / chloride ions which are released react with Ag^+ OR silver chloride / AgCl is formed	1

13. M/J 24/P22/Q4(b)

4(b)	 M1 correct dipole ($\delta^+\text{C}-\text{Br}^{\delta-}$) AND curly arrow from bond of $\text{C}-\text{Br}$ to Br (or just beyond) M2 correct intermediate ALLOW R for $-(\text{CH}_2)_5\text{CH}_3$ M3 curly arrow from lone pair on OH^- to the C of intermediate (or suitable position where $\text{C}-\text{OH}$ bond is made) <i>If $\text{S}_{\text{N}}2$ mechanism shown:</i> Apply $\text{S}_{\text{N}}2$ mechanism mark scheme only if mechanism shows OH^- / NaOH species attacking the reactant and no OH M1 correct dipole ($\delta^+\text{C}-\text{Br}^{\delta-}$) AND curly arrow from bond of $\text{C}-\text{Br}$ to Br of reactant (or just beyond) NOT M2 M3 curly arrow from lone pair on $:\text{OH}^-$ to the appropriate C of the reactant	3
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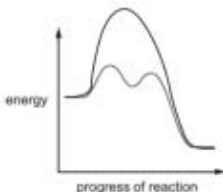
14. M/J 24/P22/Q4(d)

4(d)(i)	elimination	1
4(d)(ii)	NaOH / sodium hydroxide in ethanol	1

15. M/J 24/P23/Q4(c)

4(c)(i)	nucleophile	1
4(c)(ii)	OH^- (more reactive because) negatively charged species more attracted to halogenoalkane and / or $\text{C}^{\delta+}$ and / or intermediate and / or C^+ OR NH_3 (more reactive because) N electrons less tightly held OR N is more able to donate (lp) electrons (to make the covalent bond) OR N is less electronegative ORA	1

16. M/J 24/P23/Q4(d)

4(d)	 M1 same start energy and finishing energy AND single hump M2 transition state / activation energy shown in answer is higher than both transition state energies shown in reaction profile (as the reaction is slower)	2
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17. M/J 24/P23/Q4(e)(i)-(ii)

4(e)(i)	M1 dipole and correct arrow on C—Br of 2-bromobutane M2 lone pair on NH₃ attacking C(+) of the intermediate drawn M3 curly arrow showing bond between N—H breaking and pointing towards the N	3
4(e)(ii)	NH ₄ Br / ammonium bromide	1

18. M/J 24/P23/Q4(f)

4(f)(i)		1
4(f)(ii)	M1 2-bromo-2-methylpropane AND more stable carbocation / C⁺ M2 alkyl groups have an inductive effect (of alkyl groups) AND 2-bromo-2-methylpropane has more / most alkyl groups attached (to C⁺) / carbocation contains more alkyl groups	2

19. M/J 24/P23/Q5(b)(i)

5(b)(i)	step 1Br₂ (in the dark) step 2NaOH(aq) + heat	2
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Answer for context part (a)

5(a)	cold dilute KMnO ₄ OR cold dilute potassium manganate(VII)	1
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20. MAR 24/P22/Q2(d)(ii)

2(d)(ii)	NaOH in ethanol AND heat	1
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21. MAR 24/P22/Q4(b)(ii)

4(b)(ii)						1
	geometrical	✓	(x)	(x)	(x)	
	optical	(x)	✓	✓	✓	
CH ₃ CHBrCH ₂ CH ₃ nucleophilic substitution						

22. M/J 23/P21/Q3(e)

3(e)	CHCl ₃ + 2HF → CHClF ₂ + 2HCl	1
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23. M/J 23/P22/Q3(a)(i)

3(a)(i)	KCN in ethanol + heat	1
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24. MAR 23/P22/Q4(a)(i)

4(a)(i)	NaOH in alcohol / ethanol AND heat (under reflux)	1
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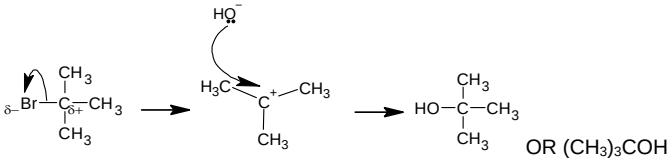
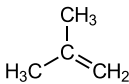
25. MAR 23/P22/Q4(b)(i)

4(b)(i)	cream(-coloured) / off-white precipitate (forms)	1
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26. MAR 23/P22/Q4(c)(i)

4(c)(i)	substitution	1
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27. O/N 22/P22/Q1(e)

1(e)(i)	 M1 correct dipole in on $\delta^+C-Br\delta^-$ (in tBuBr) AND curly arrow from $C-Br$ to Br M2 curly arrow from electron pair on O of $:OH^-$ to C^+ in carbocation M3 tBuOH product	1
1(e)(ii)	The $C-Br$ carbon is bonded to three other carbon atoms	1 <i>owtte</i>
1(e)(iii)		1

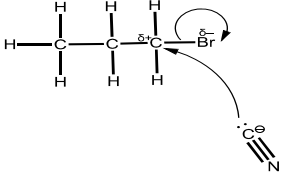
28. M/J 22/P22/Q5(a)

5(a)	reaction	reagent and conditions	type of reaction	6
	1	NaOH(aq) [M1]	substitution	
	2	NH ₃ [M2] in ethanol AND heat under pressure [M3]	substitution	
	3	NaOH in ethanol AND heat [M4]	elimination	
	M5 and M6 types of reaction [3 correct types = 2 marks and 2 correct types = 1 mark] All reagents can be identified as (correct) formula or in words			

29. M/J 22/P22/Q5(b)

5(b)	M1 C–I (covalent bond) is weaker / or a M2 lower activation energy / lower E_a (with 2-iodopropane) OR M2 <i>explain in terms of the S_N1 mechanism (that is dominant / preferred / occurring with 2-iodopropane is fast(er))</i> (identical) carbocation is made (more) quickly (with 2-iodopropane) OR low / less energy is required to make the (same) carbocation / intermediate (from 2-iodopropane) OR M2 iodine / I (of C–I) has weaker attraction of nucleus to bonding / shared pair (of electrons) due to <i>any one of</i>: • more / high shielding (of electrons from inner shells) • more / 5 electron shells in iodine • C–I greater distance from nuclei (to bonding pair of electrons)	2
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30. M/J 22/P22/Q5(c)(i)

5(c)(i)	 M1 dipole $C^{\delta+}-Br^{\delta-}$ AND curly arrow from C–Br bond to $Br^{\delta-}$ M2 lone pair on C of ^-CN AND curly arrow from lone pair to C (on C-1 of bromopropane)	2
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31. M/J 22/P23/Q4(b)(i)-(iii)

4(b)(i)	M1 hydroxy(l) M2 substitution	2
4(b)(ii)	$I^- + Ag^+ \rightarrow AgI$	1
4(b)(iii)	M AND C–I (bond) is weak(er)	1

32. M/J 22/P23/Q4(b)(v)

4(b)(v)	$CH_3CH_2CHICH_3$ M1 correct identification of I on 4C molecule M2 unambiguous structure of M – unbranched with 4C and I on 2C	2
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33. MAR 22/P22/Q3(e)(iii)

3(e)(iii)	M1: (water solvent) $CH_3CH_2CH_2OH$ AND NaBr / sodium bromide / propan-1-ol M2: (ethanol solvent) CH_3CHCH_2 AND H_2O / water / propene AND NaBr / sodium bromide	2
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